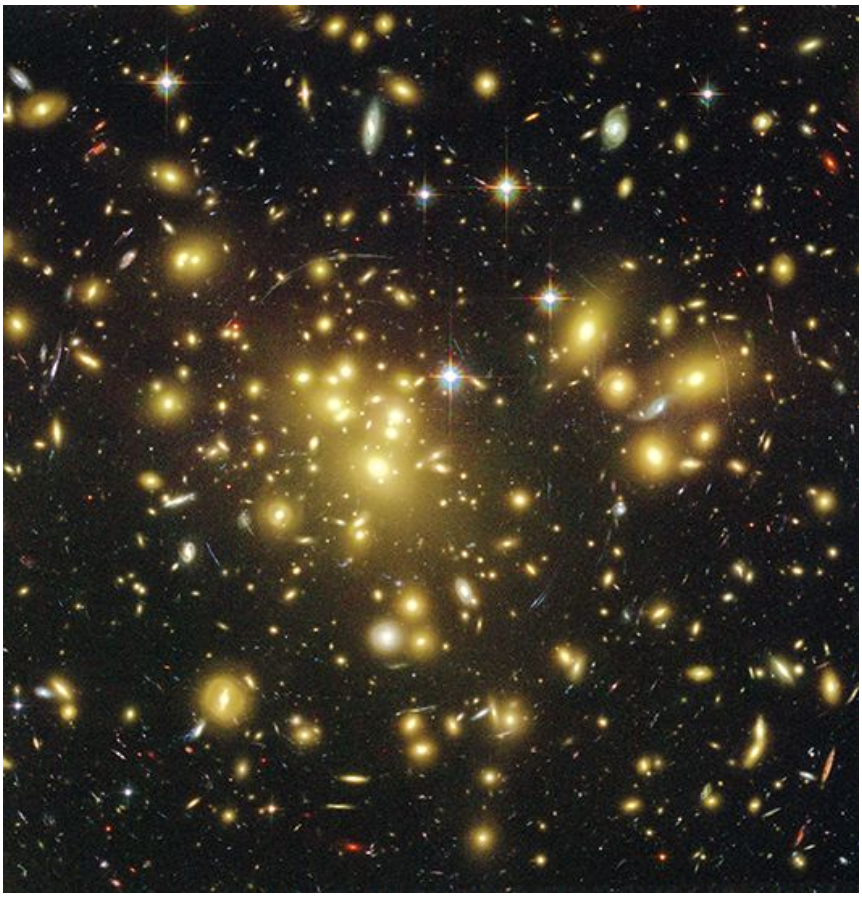


Fig. 2.1. A rich cluster of galaxies named Abell 1689 and many other more distant galaxies, as photographed by the Hubble Space Telescope.

The Earth resides in a galaxy called the Milky Way. Most of the Milky Way's stars are in the bright band of light that stretches across Earth's sky on a clear, dark night. And almost all the pinpricks of light that we see in the sky at night, not just those in the bright band, also lie in the Milky Way.

The nearest large galaxy to our own is called Andromeda (Figure 2.2). It is 2.5 million light-years from Earth. It contains about a trillion stars and is about 100,000 light-years across. The Milky Way is a sort of twin to Andromeda, about the same in size, shape, and number of stars. If Figure 2.2 were the Milky Way, then the Earth would be where I placed the yellow diamond.